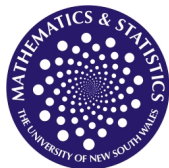




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Lecture Week 4: Decisions, Asymptotics & Importance Sampling

Instructor: Prof. Scott Sisson

Outline

1. Posterior decision making
 - Loss functions
2. Posterior asymptotics
3. Monte Carlo methods
 - Monte Carlo error
 - Importance sampling.

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1. Posterior decision making
 - Loss functions
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Summarising Posterior Information

The posterior $\pi(\theta|x)$ is a complete description of the inference about θ . In some sense $\pi(\theta|x)$ is the inference.

However, for many applications we wish to summarise this information to make a decision.

1. What is the “best” point estimate $\hat{\theta}$ of θ ?

E.g. Posterior mean, median, mode etc.

2. What decision $d^* \in \{d_1, d_2, \dots\} = \mathcal{D}$ is the optimal choice to make, given knowledge of θ ? E.g.

- How many loaves of bread to bake per day to maximise profit?
– balancing baking costs and number of loaves sold/wasted.
- How high to build sea walls to minimise cost?
– balancing construction cost, chance of breach and resulting damages.
- How much local rail infrastructure to build?
– balancing construction costs versus improvement to economy and other benefits.

Loss Functions

How do we define “best” or optimal?

Loss Function

For a decision $d \in \mathcal{D}$, a loss function

$$L(\theta, d)$$

defines the penalty in taking decision d given (fixed) parameter value θ .

- ▶ Negative loss is a gain, and is beneficial.
- ▶ Sometimes expressed as maximising utility $U(\theta, d) = -L(\theta, d)$.

The premise is then:

- ▶ Choose the decision $d^* = \arg \min_d L(\theta, d)$ that minimises the loss.

However, θ is not known, but rather $\theta \sim \pi(\theta|x)$. So alternatively

$$d^* = \arg \min_d \mathbb{E}_\pi [L(\theta, d)] = \arg \min_d \int_{\Theta} L(\theta, d) \pi(\theta|x) d\theta.$$

i.e. choose d which minimises the expected posterior loss.

Loss functions

Example: Sea Walls (with fixed θ):

Building a sea wall has initial construction cost (C) given height (h), but huge impact cost (K) in case of failure under natural disaster.

E.g.

$$L(\theta, d_h) = C(h) + K\Pr(\text{Wall exceeded in 50y}|\theta, h).$$

Decision	Height	Construction	Pr(exceeded)	Impact	$L(\theta, d_h)$
d_1	1m	\$20m	0.99	\$500m	\$515m
d_{20}	20m	\$100m	0.05	\$500m	\$125m
d_{100}	100m	\$1000m	0.0005	\$500m	\$1000.25m

Easy to see that d_{20} is optimal choice here.

But note that in practice:

1. Height would be **estimated as a continuous variable**:
 $\Rightarrow$ choose d_h for $h \in [0, \infty)$.
2. Would **minimise expected posterior loss** rather than just $L(\theta, d)$.

Loss functions

A full (decision theoretic) Bayesian setup consists of specifying:

- ▶ Prior distribution $\pi(\theta)$
- ▶ Model $f(x|\theta)$ (leading to the likelihood $L(x|\theta)$)
- ▶ Loss function $L(\theta, d)$.

Although most people only consider the first two of these.

Loss functions for estimating θ

First consider loss functions for parameter estimation:

Given $\pi(\theta|x)$ what is the optimal point estimate of θ ?

$\Rightarrow d \in \mathcal{D} = \Theta$, where $\theta \in \Theta$, so that $d = \hat{\theta}$

Consider 4 standard loss functions:

- ▶ Quadratic loss: $L(\theta, d) = (\theta - d)^2$
- ▶ Absolute error loss: $L(\theta, d) = |\theta - d|$
- ▶ 0-1 loss:

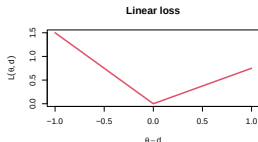
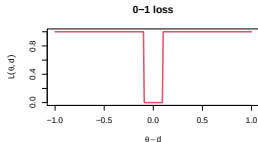
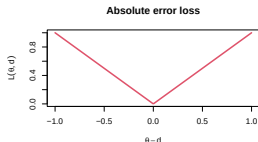
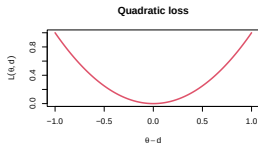
$$L(\theta, d) = \begin{cases} 0 & \text{if } |d - \theta| \leq \epsilon \\ 1 & \text{if } |d - \theta| > \epsilon \end{cases}$$

Actually, the limit of this as $\epsilon \rightarrow 0$.

- ▶ Linear loss:

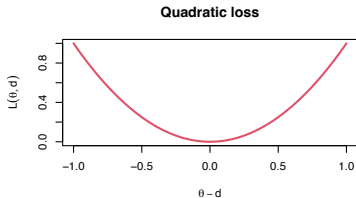
$$L(\theta, d) = \begin{cases} g(d - \theta) & \text{if } d > \theta \\ h(\theta - d) & \text{if } d < \theta \end{cases}$$

for given $g, h > 0$. ($g = h = 1$ gives Abs loss)



Quadratic Loss

$$\begin{aligned}\mathbb{E}_{\pi}[L(\theta, d)] &= \int L(\theta, d)\pi(\theta|x)d\theta \\&= \int (\theta - d)^2\pi(\theta|x)d\theta \\&= \int (\theta - \mathbb{E}(\theta|x) + \mathbb{E}(\theta|x) - d)^2\pi(\theta|x)d\theta \\&= \int (\theta - \mathbb{E}(\theta|x))^2\pi(\theta|x)d\theta + \int (\mathbb{E}(\theta|x) - d)^2\pi(\theta|x)d\theta \\&\quad + 2 \int (\theta - \mathbb{E}(\theta|x))(\mathbb{E}(\theta|x) - d)\pi(\theta|x)d\theta \\&= \text{Var}(\theta|x) + (\mathbb{E}(\theta|x) - d)^2 + 0\end{aligned}$$



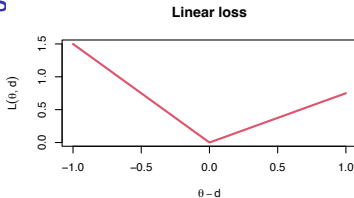
- ▶ This is minimised when $d = \mathbb{E}(\theta|x)$.
- ▶ The posterior mean minimises quadratic loss.
- ▶ The minimum expected loss is the posterior variance.

Linear Loss

Linear loss:

$$L(\theta, d) = \begin{cases} g(d - \theta) & \text{if } d > \theta \\ h(\theta - d) & \text{if } d < \theta \end{cases}$$

for given $g, h > 0$.



Let q be the $\frac{h}{g+h}$ quantile of the posterior distribution i.e.

$$\frac{h}{g+h} = \int_{-\infty}^q \pi(\theta|x) d\theta \quad (= P_{\pi}(\theta \leq q)).$$

Suppose (without loss of generality) that $q < d$. Then

$$L(\theta, q) - L(\theta, d) = \begin{cases} g(q - d) & \text{if } \theta \leq q < d \\ (g+h)\theta - hq - gd & \text{if } q < \theta < d \\ h(d - q) & \text{if } q < d \leq \theta \end{cases}$$

as

- ▶ $L(\theta, q) - L(\theta, d) = g(q - \theta) - g(d - \theta) = g(q - d)$
- ▶ $L(\theta, q) - L(\theta, d) = h(\theta - q) - g(d - \theta) = (g+h)\theta - hq - gd$
- ▶ $L(\theta, q) - L(\theta, d) = h(\theta - q) - h(\theta - d) = h(d - q)$

Linear Loss

Now, for $q < \theta < d$,

$$(g + h)\theta - hq - gd < h(d - q) \quad (\text{put } \theta = d \text{ to maximise LHS})$$

which means that

$$L(\theta, q) - L(\theta, d) \leq \begin{cases} g(q - d) & \text{if } \theta \leq q \\ h(d - q) & \text{if } q < \theta. \end{cases}$$

Now,

$$\begin{aligned} \mathbb{E}[L(\theta, q) - L(\theta, d)] &\leq g(q - d)\frac{h}{g + h} + h(d - q)(1 - \frac{h}{g + h}) \\ &= \frac{1}{g + h} [gh(q - d) + gh(d - q)] = 0. \end{aligned}$$

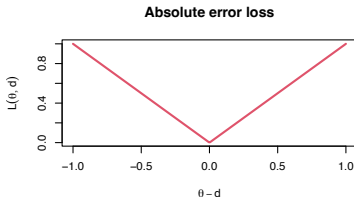
i.e. $\mathbb{E}[L(\theta, q)] \leq \mathbb{E}[L(\theta, d)]$ for all d .

So linear loss is minimised at $d = q$, the $\frac{h}{g+h}$ posterior quantile.

Absolute Error Loss

Absolute error loss:

$$L(\theta, d) = |\theta - d|$$



Absolute error loss = Linear loss with $g = h = 1$.

When $g = h = 1$, q is the median of the posterior distribution.

⇒ The posterior median minimises absolute error loss.

0-1 Loss

0-1 loss:

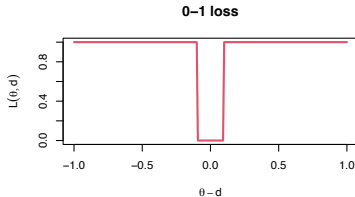
$$L(\theta, d) = \begin{cases} 0 & \text{if } |d - \theta| \leq \epsilon \\ 1 & \text{if } |d - \theta| > \epsilon \end{cases}$$

as $\epsilon \rightarrow 0$.

Here

$$\begin{aligned} \mathbb{E}[L(\theta, d)] &= 0 \times \Pr(|d - \theta| \leq \epsilon) + 1 \times \Pr(|\theta - d| > \epsilon) \\ &= \Pr(|\theta - d| > \epsilon) \\ &= 1 - \Pr(|\theta - d| \leq \epsilon). \end{aligned}$$

- ▶ $|\theta - d| \leq \epsilon$ defines an interval $[d - \epsilon, d + \epsilon]$ of length 2ϵ on $\pi(\theta|x)$
- ▶ d can be considered a “location parameter”
- ▶ **To minimise**, choose interval with highest probability
i.e. high density region
- ▶ Then d is mid-point of interval with highest probability
- ▶ Choosing ϵ arbitrarily small **will select the posterior mode for d .**



→ The posterior mode minimises 0-1 loss

Loss functions for making other decisions

Example: Baking loaves of bread

- ▶ c = the cost of baking a loaf of bread
- ▶ $s > c$ is price loaf sells for
- ▶ $\pi(d|x)$ is the (posterior) distribution of demand for bread
- ▶ $b \in \{0, 1, \dots\}$ is the decision (i.e. number of loaves to bake).
- ▶ Possible profit model: If baker bakes b loaves with demand d then profit is

$$\text{Profit} = \begin{cases} (s - c)d - c(b - d) & \text{for } b > d \\ (s - c)b & \text{for } b = d \\ (s - c)b & \text{for } b < d \end{cases}$$

Can combine the last two as Profit = $(s - c)b$ for $b \leq d$.

Loss functions for making other decisions

Want to minimise expected loss (real and opportunity loss)

- ▶ Make Profit relative to selling $b = d$ loaves
(i.e. subtract $(s - c)d$ from all terms)

$$\text{Profit}' = \begin{cases} -c(b - d) & \text{for } b > d \\ (s - c)b - (s - c)d = (s - c)(b - d) & \text{for } b \leq d \end{cases}$$

- ▶ Loss function ($= -\text{Profit}'$):

$$L(b, d) = \begin{cases} c(b - d) & \text{for } b > d \quad (\text{cost of baking surplus loaves}) \\ (s - c)(d - b) & \text{for } b \leq d \quad (\text{lost profit from not baking enough loaves}) \end{cases}$$

- ▶ This is linear loss with $g = c$ and $h = s - c$
- ▶ Therefore the optimal decision minimising expected posterior loss is the $b^* = \frac{h}{g+h} = \frac{s-c}{s}$ quantile of $\pi(d|x)$.

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 - Loss functions
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Posterior Asymptotics

What happens to the posterior distribution as $n \rightarrow \infty$?

Maximum likelihood estimation (recall):

- ▶ Consistency: $\hat{\theta}_{MLE} \rightarrow \theta_0$
- ▶ Asymptotic normality: $\hat{\theta}_{MLE} \sim N(\theta_0, I_n(\theta_0)^{-1})$

1. Consistency (concentration):

If the “true” value of $\theta = \theta_0$,
and if $\pi(\theta_0) \neq 0$ (**or is non-zero in a neighbourhood of θ_0**),
then with increasing amounts of data ($n \rightarrow \infty$) the posterior
probability that θ equals (or lies in a neighbourhood of) $\theta_0 \rightarrow 1$.

Property akin to classical notion of ‘consistency’

2. Asymptotic Normality:

As $n \rightarrow \infty$ then

$$\pi(\theta|x) \rightarrow N(\theta_0, I_n(\theta_0)^{-1}).$$

All the following arguments are heuristic/outline proofs.

Assume: the model is well specified ($f(x|\theta)$ is the true data generation process for some value of $\theta = \theta_0$).

Posterior Asymptotics - Consistency

1. Consistency:

Let $x_1, \dots, x_n \sim f(x|\theta_0)$ be *iid* observations.

Then the posterior is

$$\begin{aligned}\pi(\theta|x_1, \dots, x_n) &\propto \pi(\theta) \prod_{i=1}^n f(x_i|\theta) \\ &= \pi(\theta) \exp\left(\sum_{i=1}^n \log f(x_i|\theta)\right) \\ &= \pi(\theta) \exp(n\ell_n(\theta))\end{aligned}$$

where $\ell_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log f(x_i|\theta)$.

Posterior Asymptotics - Consistency

From law of large numbers, we have for fixed θ that as $n \rightarrow \infty$,

$$\ell_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log f(x_i|\theta) \rightarrow \mathbb{E}[\log f(x|\theta)]$$

where

$$\mathbb{E}[\log f(x|\theta)] = \int [\log f(x|\theta)] f(x|\theta) dx. \quad (1)$$

$$\begin{aligned} \text{Also: } \mathbb{E}[\ell_n(\theta)] &= \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n \log f(x_i|\theta)\right] = \frac{1}{n} \times n \mathbb{E}[\log f(x|\theta)] \\ &= \mathbb{E}[\log f(x|\theta)] \quad \forall n. \end{aligned}$$

Therefore we make the following heuristic approximation for large n ,

$$\frac{1}{n} \sum_{i=1}^n \log f(x_i|\theta) \approx \mathbb{E}[\ell_n(\theta)]$$

so that

$$\pi(\theta|x_1, \dots, x_n) \propto \pi(\theta) \exp(n \mathbb{E}[\ell_n(\theta)])$$

Posterior Asymptotics - Consistency

Note that from (1) we have

$$\arg \max_{\theta} \mathbb{E}[\ell_n(\theta)] = \theta_0. \quad (2)$$

Hence, for $\theta \neq \theta_0$, it follows that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\pi(\theta_0 | x_1, \dots, x_n)}{\pi(\theta | x_1, \dots, x_n)} &= \lim_{n \rightarrow \infty} \frac{\pi(\theta_0) \exp\{n\ell_n(\theta_0)\}}{\pi(\theta) \exp\{n\ell_n(\theta)\}} \\ &\approx \lim_{n \rightarrow \infty} M \frac{\exp\{n\mathbb{E}[\ell_n(\theta_0)]\}}{\exp\{n\mathbb{E}[\ell_n(\theta)]\}} \\ &= \lim_{n \rightarrow \infty} M \exp\{n(\mathbb{E}[\ell_n(\theta_0)] - \mathbb{E}[\ell_n(\theta)])\} = \infty \end{aligned}$$

with probability 1, for some $M > 0$ (not necessarily finite), since (2) implies $(\mathbb{E}[\ell_n(\theta_0)] - \mathbb{E}[\ell_n(\theta)]) \geq 0$ for all θ .

This is sufficient to prove consistency, **provided that $\pi(\theta_0) \neq 0$** .

(This also assumes an identifiable model – expectation needs extra assumptions for non-identifiability)

Posterior Asymptotics - Asymptotic Normality

2. Asymptotic Normality: (univariate and continuous θ)

Recall: Taylor series

For a function $f(x)$ that is infinitely differentiable at a , then

$$f(x) = f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!} + \dots$$

Now expand $\mathbb{E}[\ell_n(\theta)]$ as a Taylor series about $a = \theta_0 \dots$

$$f(x) = \mathbb{E}[\ell_n(\theta)] \quad (x = \theta)$$

$$\begin{aligned}\mathbb{E}[\ell_n(\theta)] &\approx \mathbb{E}[\ell_n(\theta_0)] + \mathbb{E}'[\ell_n(\theta_0)](\theta - \theta_0) + \frac{1}{2}\mathbb{E}''[\ell_n(\theta_0)](\theta - \theta_0)^2 \\ &= \mathbb{E}[\ell_n(\theta_0)] - \frac{1}{2v}(\theta - \theta_0)^2\end{aligned}$$

Posterior Asymptotics- Asymptotic Normality

since

$$\begin{aligned}\mathbb{E}'[\ell_n(\theta_0)] &= \left. \frac{\partial}{\partial \theta} \int [\log f(x|\theta)] f(x|\theta_0) dx \right|_{\theta=\theta_0} \\ &= \left. \int \frac{\partial}{\partial \theta} [\log f(x|\theta)] f(x|\theta_0) dx \right|_{\theta=\theta_0} = 0\end{aligned}$$

as the derivative of the log likelihood must be zero at $\theta = \theta_0$, and where

$$v = -1/\mathbb{E}''[\ell_n(\theta_0)] \quad (= I(\theta)^{-1})$$

$$\begin{aligned}\text{and where } -\mathbb{E}''[\ell_n(\theta)] &= -\frac{\partial^2}{\partial \theta^2} \int [\log f(x|\theta)] f(x|\theta_0) dx \\ &= -\int \frac{\partial^2 \log f(x|\theta)}{\partial \theta^2} f(x|\theta_0) dx = I(\theta).\end{aligned}$$

Posterior Asymptotics- Asymptotic Normality

If prior density $\pi(\theta)$ is continuous at θ_0 and $\pi(\theta_0) > 0$, then $\pi(\theta)$ will be approximately constant in an increasingly small neighbourhood of θ_0 (consistency) Hence, for large n , we have the approximation

$$\begin{aligned}\pi(\theta|x_1, \dots, x_n) &\propto \exp\left(n\mathbb{E}[\ell_n(\theta_0)] - \frac{n}{2v}(\theta - \theta_0)^2\right) \\ &\propto \exp\left(-\frac{n}{2v}(\theta - \theta_0)^2\right).\end{aligned}$$

i.e.

$$\theta|x \sim N(\theta_0, I_n(\theta_0)^{-1}).$$

Note: $nI(\theta) = I_n(\theta)$ the Fisher information based on the likelihood.

So for large n , the posterior distribution is approximately Normal, centred on the true value of θ , with variance $I_n(\theta_0)^{-1}$.

Notes:

- ▶ This occurs independently of the prior specification (i.e. **any prior**), provided that $\pi(\theta_0) \neq 0$ (or is non-zero in a neighbourhood of θ_0).
- ▶ As we don't know θ_0 in practice, **often substitute with MLE $\hat{\theta}$, as $\lim_{n \rightarrow \infty} \hat{\theta} = \theta_0$.**

Posterior Asymptotics

Example: Normal model

Let $x_1, \dots, x_n \sim N(\theta, \sigma^2)$ where σ^2 is known.

As usual, this gives the likelihood

$$f(x|\theta) \propto \exp \left\{ -\frac{\sum_{i=1}^n (x_i - \theta)^2}{\sigma^2} \right\}$$

from which we obtain

$$\log(f(x|\theta)) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \theta)^2 \quad \frac{d \log(f(x|\theta))}{d\theta} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \theta)$$

and so

$$\frac{d^2 \log(f(x|\theta))}{d^2\theta} = -n/\sigma^2.$$

Hence, the mle is $\hat{\theta} = \bar{x}$ and $I_n(\theta) = n/\sigma^2$. So asymptotically, as $n \rightarrow \infty$,

$$\theta|x \sim N(\bar{X}, \sigma^2/n).$$

This is true for any prior distribution which places non-zero probability around the true value of θ .

Likelihood Asymptotics

Consider again the likelihood model $x \sim \text{Bin}(n, \theta)$.

$$f(x|\theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}, \quad x = 0, \dots, n$$

thus

$$\log(f(x|\theta)) \propto x \log \theta + (n - x) \log(1 - \theta)$$

so,

$$\frac{d \log(f(x|\theta))}{d\theta} = \frac{x}{\theta} - \frac{n - x}{1 - \theta}$$

and

$$\frac{d^2 \log f(x|\theta)}{d^2\theta} = -\frac{x}{\theta^2} - \frac{n - x}{(1 - \theta)^2}.$$

Consequently

$$I_n(\theta) = \frac{n\theta}{\theta^2} + \frac{n(1 - \theta)}{(1 - \theta)^2} = \frac{n}{\theta(1 - \theta)}. \quad (\mathbb{E}(X) = n\theta)$$

Now $\hat{\theta} = x/n$ and thus as $n \rightarrow \infty$, we have

$$\theta|x \sim N\left(\frac{x}{n}, \frac{\frac{x}{n}(1 - \frac{x}{n})}{n}\right).$$

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Monte Carlo error

Recall: We want to estimate an expectation:

$$\int_{\mathcal{X}} h'(x) dx = \int_0^1 h(u) du \approx \frac{1}{N} \sum_{i=1}^N h(u^{(i)}) = \bar{h}$$

We know that the variability of this estimator goes down with N .
How can we understand this variability and control it?

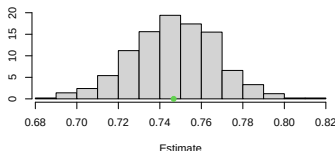
E.g. $\int_0^1 \exp(-u^2) du$

Option 1: Repeated simulation (easy, but poor)

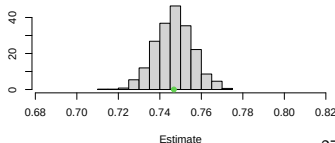
- ▶ Draw N samples $u^{(1)}, \dots, u^{(N)} \sim U(0, 1)$
- ▶ Compute $\bar{h}$
- ▶ Repeat N_{rep} times (here $N_{rep} = 1,000$)
- ▶ Explore the variability.

- ✓ Variability easy to obtain
- ✗ Far too much computation!

Estimator distribution: N = 100



Estimator distribution: N = 500



Monte Carlo error

Recall: We want to estimate an expectation:

$$\mu_{true} = \int_{\mathcal{X}} h'(x) dx = \int_0^1 h(u) du \approx \frac{1}{N} \sum_{i=1}^N h(u^{(i)}) = \bar{h}$$

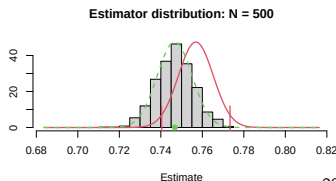
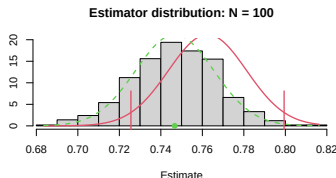
We know that the variability of this estimator goes down with N .
How can we understand this variability and control it?

E.g. $\int_0^1 \exp(-u^2) du$

Option 2: Exploit the CLT (easy, and good)

- ▶ Draw N samples $u^{(1)}, \dots, u^{(N)} \sim U(0, 1)$
- ▶ Compute $h(u^{(i)})$: then $\bar{h}$, and s_h (st. dev.)
- ▶ Note that $\bar{h} \sim N(\mu_{true}, \frac{s_h^2}{N})$ for large N
- ▶ Use this to make calculations for N .

- ✓ Easy to compute
- ✓ Trivial computation!



Monte Carlo error

- ▶ Note that $\bar{h} \sim N(\mu_{true}, \frac{s_h^2}{N})$ for large N
- ▶ Use this to make calculations for N .

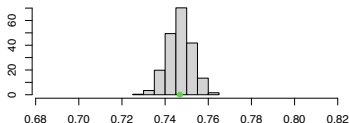
Can now use this to choose N so that estimator variability is controlled.
Standard “sample size calculation” process.

E.g. Want to estimate μ_{true} to within E with probability $1 - \alpha$.
How many samples N ?

- ▶ Draw N samples $u^{(1)}, \dots, u^{(N)} \sim U(0, 1)$
- ▶ Compute $h(u^{(i)})$: then $\bar{h}$, and s_h (st. dev.)
- ▶ Want N such that $z_{1-\alpha/2} \frac{s_h}{\sqrt{N}} \leq E$
- ▶ I.e. take: $N \geq \left(\frac{z_{1-\alpha/2} s_h}{E} \right)^2$. Here with $E = 0.01$, $N \geq 1,356$

- ✓ Works for independent samples.
- !! More complicated if samples are dependent (e.g. MCMC – see later).

Estimator distribution: N = 1360



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Recap: Rejection Sampling

We want to sample from a distribution $f(x)$, but this is difficult to do.

Rejection sampling approach:

- ▶ Sample from a distribution, $x^* \sim g(x)$, from which it is easy to simulate
- ▶ Accept x^* with some probability, p .

If we choose p correctly, we can make accepted samples come from $f(x)$.

Example:

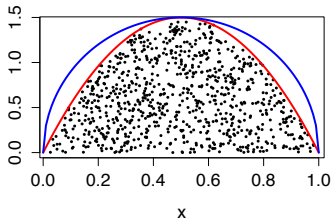
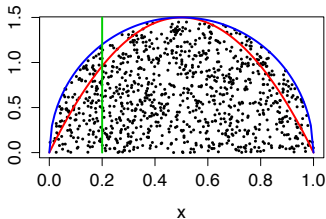
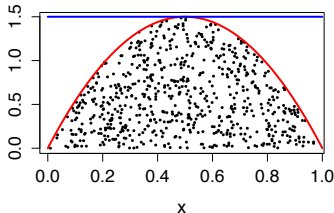
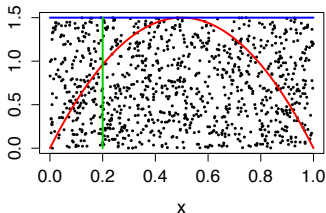
Obtain N samples from $\text{Beta}(2, 2)$.

Solution:

Inversion sampling is inefficient as requires numerical inversion of $F(x)$.
Instead, try rejection sampling.

Recap: Rejection Sampling

Example: Obtain N samples from $\text{Beta}(2, 2)$.



red = $f(x)$, blue = $Kg(x)$

Top: $g(x) = \text{Beta}(1, 1)$, efficiency = $681/1000 = 0.681$

Bottom: $g(x) = \text{Beta}(1.5, 1.5)$, efficiency = $848/1000 = 0.848$

Recap: Rejection Sampling

Algorithm version 1

1. Generate x^* from $g(x)$
2. Generate y^* from $U(0, Kg(x^*))$
3. Accept x^* if $y^* \leq f(x^*)$

Algorithm version 2

1. Generate x^* from $g(x)$
2. Accept x^* with probability $\frac{f(x^*)}{Kg(x^*)}$

Importance sampling

Consider a variation on Version 2:

- ▶ We keep all samples from $g(x)$.
- ▶ But instead weight them, so that they are samples from $f(x)$.

Importance Sampling

Importance Sampling Algorithm

For $i = 1, \dots, N$:

1. Generate $x^{(i)}$ from $g(x)$
2. Give $x^{(i)}$ a weight $w^{(i)} \propto \frac{f(x^{(i)})}{g(x^{(i)})}$

Notes:

- ▶ The samples $(x^{(1)}, w^{(1)}), \dots, (x^{(N)}, w^{(N)})$ are weighted samples from $f(x)$.
- ▶ Weight is $\propto f(x)/g(x)$ not $f(x)/Kg(x)$ (as for rejection sampling) as K is lost in proportionality

How does inference work for weighted samples?

Importance Sampling

How does inference work for weighted samples? (Assume $\int f(x)dx = 1$.)

Unweighted expectation:

$$\mathbb{E}_g[h(x)] = \int h(x)g(x)dx \approx \frac{1}{N} \sum_{i=1}^N h(x^{(i)})$$

where $x^{(1)}, \dots, x^{(N)}$ are samples from $g(x)$.

Weighted expectation: Defining weights $w(x) = f(x)/g(x)$, then

$$\begin{aligned}\mathbb{E}_g[w(x)h(x)] &= \int w(x)h(x)g(x)dx \approx \frac{1}{N} \sum_{i=1}^N w^{(i)}h(x^{(i)}) \\ &= \int h(x)f(x)dx \\ &= \mathbb{E}_f[h(x)]\end{aligned}$$

where $x^{(1)}, \dots, x^{(N)}$ are samples from $g(x)$.

i.e. Weighted expectations under $g(x)$ act as expectations under $f(x)$.

Importance Sampling

What if $f(x)$ is unnormalised?

We then have $f(x) = \tilde{f}(x)/Z$ where $Z = \int \tilde{f}(x)dx$ is unknown.

Weighted expectation: Defining weights $\tilde{w}(x) = \tilde{f}(x)/g(x)$, note that:

$$\mathbb{E}_g[\tilde{w}(x)] = \int \tilde{w}(x)g(x)dx = \int \tilde{f}(x)dx = Z \approx \frac{1}{N} \sum_{i=1}^N \tilde{w}(x^{(i)}).$$

$$\begin{aligned}\mathbb{E}_f[h(x)] &= \int h(x)f(x)dx = \frac{1}{Z} \int h(x)\tilde{f}(x)dx \\ &= \frac{1}{Z} \int \tilde{w}(x)h(x)g(x)dx = \frac{\mathbb{E}_g[\tilde{w}(x)h(x)]}{\mathbb{E}_g[\tilde{w}(x)]} \\ &\approx \frac{\frac{1}{N} \sum_{i=1}^N \tilde{w}(x^{(i)})h(x^{(i)})}{\frac{1}{N} \sum_{i=1}^N \tilde{w}(x^{(i)})} = \sum_{i=1}^N W(x^{(i)})h(x^{(i)})\end{aligned}$$

where $W(x^{(i)}) = \tilde{w}(x^{(i)}) / \sum_{j=1}^N \tilde{w}(x^{(j)})$ and $x^{(1)}, \dots, x^{(N)} \sim g(x)$.

i.e. normalise weights (to sum to one) then take expectations.

* – this is a biased estimator for finite N (but is asymptotically unbiased, i.e. as $N \rightarrow \infty$).

Importance Sampling

Technical note: a biased expectation:

$$\frac{\mathbb{E}_g[\tilde{w}(x)h(x)]}{\mathbb{E}_g[\tilde{w}(x)]} \approx \frac{\frac{1}{N} \sum_{i=1}^N \tilde{w}(x^{(i)})h(x^{(i)})}{\frac{1}{N} \sum_{i=1}^N \tilde{w}(x^{(i)})} \quad x^{(i)} \sim g^*$$

* – this is a biased estimator for finite N (but is asymptotically unbiased, i.e. as $N \rightarrow \infty$).

$\frac{1}{N} \sum_{i=1}^N \tilde{w}(x^{(i)})h(x^{(i)})$ is an unbiased estimator of $\mathbb{E}_g[\tilde{w}(x)h(x)]$ ✓

$\frac{1}{N} \sum_{i=1}^N \tilde{w}(x^{(i)})$ is an unbiased estimator of $\mathbb{E}_g[\tilde{w}(x)]$ ✓

BUT $\frac{\frac{1}{N} \sum_{i=1}^N \tilde{w}(x^{(i)})h(x^{(i)})}{\frac{1}{N} \sum_{i=1}^N \tilde{w}(x^{(i)})}$ is NOT an unbiased estimator of $\frac{\mathbb{E}_g[\tilde{w}(x)h(x)]}{\mathbb{E}_g[\tilde{w}(x)]}$ ✗

Why? Because

$$\frac{\mathbb{E}(X)}{\mathbb{E}(Y)} \neq \mathbb{E}\left(\frac{X}{Y}\right).$$

However, these are equal asymptotically, as $N \rightarrow \infty$.

Importance Sampling

Summary: Normalised vs unnormalised importance sampling (IS).

Unnormalised IS: (can use when $Z = \int \tilde{f}(x)dx$ known)

$$\mathbb{E}_f[h(x)] \approx \frac{1}{N} \sum_{i=1}^N w(x^{(i)})h(x^{(i)}), \quad w(x^{(i)}) = f(x^{(i)})/g(x^{(i)}), \quad x^{(i)} \sim g$$

Unnormalised estimator is unbiased,

i.e. $\mathbb{E}_g \left[\frac{1}{N} \sum_{i=1}^N w(x^{(i)})h(x^{(i)}) \right] = \mathbb{E}_f[h(x)]$ for all N .

Normalised IS: (use either when Z is known or unknown)

$$\mathbb{E}_f[h(x)] \approx \sum_{i=1}^N W(x^{(i)})h(x^{(i)}), \quad W(x^{(i)}) = \frac{\tilde{w}(x^{(i)})}{\sum_{j=1}^N \tilde{w}(x^{(j)})}, \quad x^{(i)} \sim g$$

Normalised estimator is biased for finite N but asymptotically unbiased,

i.e. $E_g \left[\sum_{i=1}^N W(x^{(i)})h(x^{(i)}) \right] \rightarrow \mathbb{E}_f[h(x)]$ as $N \rightarrow \infty$.

Importance Sampling

Example:

Simulate from the density

$$f(x) = \begin{cases} 20x(1-x)^3 & 0 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Use importance sampling with

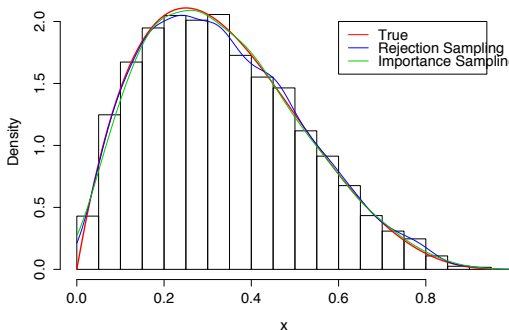
$$g(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(that is, use the uniform density on $[0, 1]$).

Previously (Lecture 3) used rejection sampling with

- ▶ $K = 135/64$ (if $f(x)$ is normalised as $f(x) = 20x(1-x)^3$)
- ▶ $K = 27/256$ (if $f(x) \propto x(1-x)^3$).

Importance Sampling

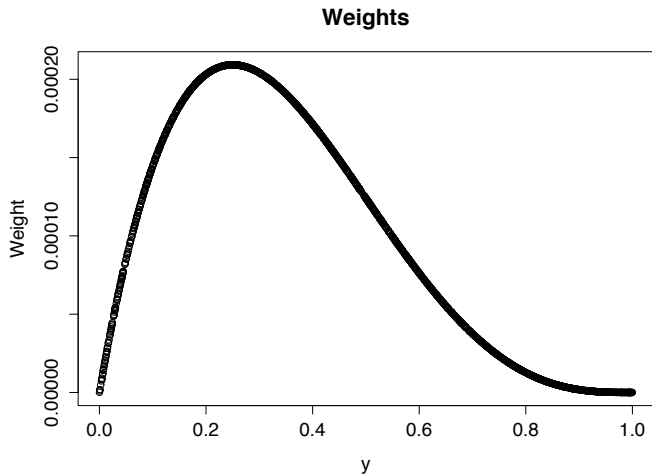


```
# Rejection sampling
L=10000
K=135/64
x=runif(L)
ind=(runif(L)<(20*x*(1-x)^3/K))
hist(x[ind],probability=T,
     xlab="x",ylab="Density",main="")
xx=seq(0,1,length=100)
lines(xx,20*xx*(1-xx)^3,lwd=2,col=2)
d=density(x[ind],from=0,to=1)
lines(d,col=4)
```

```
# Importance sampling
y=runif(L)
w=20*y*(1-y)^3
wTilde=y*(1-y)^3 # = tilde(f)/g
W=wTilde/sum(wTilde)
d=density(y,weights=W,from=0,to=1)
lines(d,col=3)
```

```
> mean(x[ind])
[1] 0.3353401
> mean(y)
[1] 0.5042156
> mean(w*y)
[1] 0.3334709
> sum(W*y)
[1] 0.3363843
```

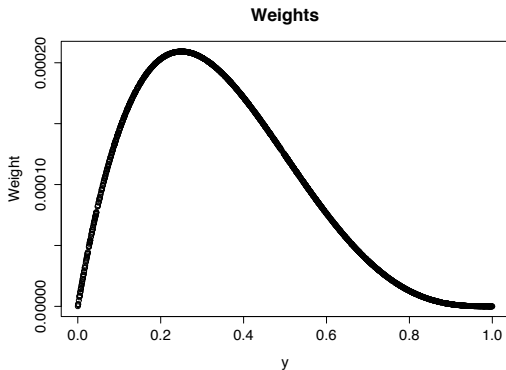
Importance Sampling



- ▶ Shows density of $f(x)$ only as $g(x)$ is uniform.
- ▶ In general shows $\propto f(x)/g(x)$

Importance Sampling

Note:



- ▶ Variability in weights mean some weighted samples contribute more than others in computations
- ▶ Samples with low weights have small contribution
- ▶ $\Rightarrow$ for efficiency, would like all samples to contribute as equally as possible

Concept: Variability of weights

- ▶ If weights are highly variable then $\text{Var}(w^{(i)})$ is high. (=bad)
- ▶ If weights have low variability then they are all similar values ($\text{Var}(w^{(i)})$ is small) (=good).
- ▶ For best performance, prefer low variability weights

Importance Sampling

Weights variance usually measured through **Effective Sample Size (ESS)**

$$\text{ESS} = \left[\sum_{i=1}^N (W^{(i)})^2 \right]^{-1}$$

where $W^{(i)} = W(x^{(i)})$ are normalised weights.

Note that

$$1 \leq \text{ESS} \leq N.$$

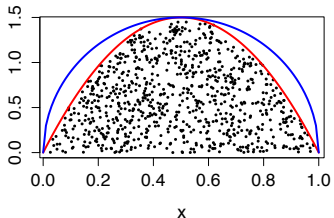
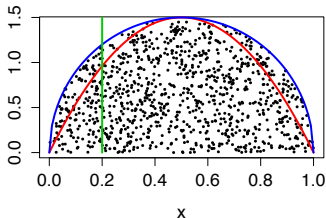
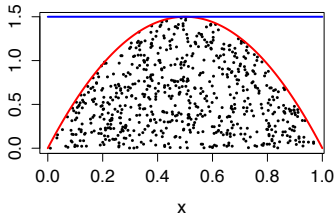
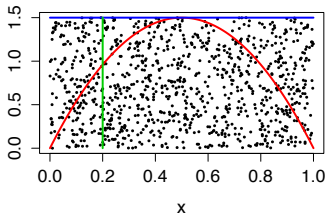
- ▶ $\text{ESS}=1$ when $W^{(1)} = 1, W^{(2)}, \dots, W^{(N)} = 0$ (sample depletion)
- ▶ $\text{ESS}=N$ when $W^{(i)} = 1/N$ for all i .
- ▶ Loose interpretation: **equivalent number of equally weighted independent samples**

To maximise ESS, **choose $g(x)$ to closely match $f(x)$.**

Same idea when improving efficiency of rejection sampling.

Importance Sampling

Example: Obtain N samples from $\text{Beta}(2, 2)$.



red = $f(x)$, blue = $g(x)$

Top: $g(x) = \text{Beta}(1, 1)$, eff = 681/1000 ESS=4172.82 (84%)

Bottom: $g(x) = \text{Beta}(1.5, 1.5)$, eff = 848/1000 ESS=4634.02 (93%)

Importance Sampling

In order to estimate the integral of $Z = \int f(x)dx$ we can:

1. Reparameterise to an integral over $[0, 1]$ then compute as

$$\int_{\mathcal{X}} f(x)dx = \int_0^1 f'(u)du \approx \frac{1}{N} \sum_i f'(u^{(i)})$$

with $u^{(1)}, \dots, u^{(N)} \sim U(0, 1)$ (as before).

2. As above, but with $u^{(1)}, \dots, u^{(N)} \sim g(u)$ on $(0, 1)$.

$$\int_{\mathcal{X}} f(x)dx = \int_0^1 f'(u)du = \int_0^1 \frac{f'(u)}{g(u)} g(u)du \approx \frac{1}{N} \sum_i \frac{f'(u^{(i)})}{g(u^{(i)})}$$

where $g(u) = U(0, 1)$ recovers the first method.

3. Integrate without transformation to $[0, 1]$

$$\int_{\mathcal{X}} f(x)dx = \int_{\mathcal{X}} \frac{f(x)}{g(x)} g(x)dx \approx \frac{1}{N} \sum_i \frac{f(x^{(i)})}{g(x^{(i)})}$$

where $x^{(1)}, \dots, x^{(N)} \sim g(x)$.

Aim to choose g to minimise $\text{Var}(f/g)$ e.g. by maximising ESS.